

Lesson 5.4: Utility analysis

Risk attitudes and utility

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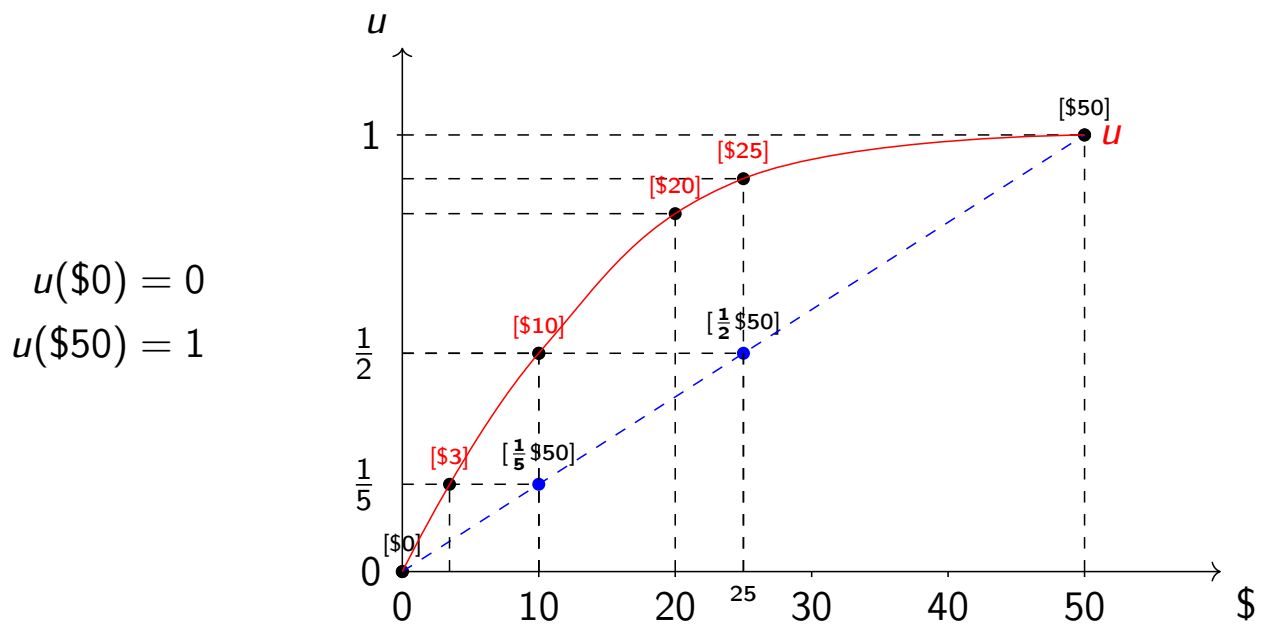
Utility values

Alice would only bet \$10 if she believed her chances of winning \$50 were at least one in two.

- ▶ Alice's certainty equivalent for (lottery) *one-in-two chance of winning \$50* is \$10
- ▶ $u(\$10) = u(\left[\frac{1}{2} : \$50 \mid \frac{1}{2} : \$0\right])$
- ▶ how do we measure overall preference (utility)?
- ▶ convention: utility scale based on worst/best outcomes:

outcome	utility
\$0	0
\$50	1

Utility curves



Reference lotteries lie on diagonal:

$$u\left(\left[\frac{1}{2} : \$50 \mid \frac{1}{2} : \$0\right]\right) = \frac{1}{2}u(\$50) + \frac{1}{2}u(\$0) = \frac{1}{2}$$

$$u([p : \$50 \mid (1-p) : \$0]) = p$$

The geometry of risk preference

- Relation between certainty equivalent and fair value determines risk attitude
- Shape of utility curve determines decision-maker's risk attitude:

utility curve shape		risk attitude
concave down	→	risk averse
concave up	→	risk seeking
linear	→	risk neutral